A few examples of acids and their corresponding conjugate bases are as follows.

Acids		Protons		Conjugate Bases
H_2O	$\rightarrow$	H^+	+	OH^-
HCl	$\rightarrow$	H^+	+	Cl^-
H_2CO_3	$\rightarrow$	H^+	+	HCO_3^-
CH_3COOH	$\rightarrow$	H^+	+	CH_3COO^-
NH_4^+	$\rightarrow$	H^+	+	NH_3

Some metallic hydroxides such as NaOH and KOH are commonly referred to as alkalies. They dissociate to form metallic ion and OH^- ion. The latter, being a base, accepts H^+ ions.

The substances which can function both as acids and bases are referred to as ampholytes. Water is the best example for an ampholyte. Similarly, amino acids are also ampholytes.

Acids and bases in biological systems

In general, acids are produced in the body as the end products of many metabolic reactions. These include the volatile acids like carbonic acid (most predominant, about 20,000 mEq/day) or non-volatile acids (about 80 mEq/day) such as lactic acid, sulphuric acid, phosphoric acid etc. Carbonic acid is formed from the metabolic product CO_2; lactic acid is produced in anaerobic metabolism; sulphuric acid is generated from proteins (sulfur containing amino acids); phosphoric acid is derived from organic phosphates (e.g. phospholipids). All these acids add up H^+ ions to the blood.

The formation of bases in the body under normal circumstances is negligible. Some amount of bicarbonate is generated from carbondioxide. The ammonia produced from amino acids is converted to urea.

A diet rich in animal proteins (non-vegetarian diet) results in more acid production by the body that ultimately leads to the excretion of urine which is profoundly acidic. A vegetarian diet has a tendency for a net production of bases. This is due to the fact that vegetarian diet produces salts of organic acids such as sodium lactate which can utilize H^+ ions produced in the body. For this reason, a vegetarian diet has an alkalizing effect on the body. This is reflected by the excretion of neutral or slightly alkaline urine.

Unit 1.indd 26 21-05-2018 15:07:39